

PRIVACY POLICY ANALYZER (TEXT BASED)

Team Members:

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Statement:

9. Privacy Policy Analyzer (Text-based)

- **Project Goal:** Create an application that takes a privacy policy as text input and highlights keywords or phrases related to data collection, sharing, and user rights.
- **Features:** Text input, keyword highlighting, summary generation.
- **Database:** Store a dictionary of privacy-related terms and their classifications.
- **OOP:** Classes for `TextAnalyzer`, `KeywordMatcher`.

Execution:

```
sudhan@DESKTOP-THAM0AM x + v
[sudhan@DESKTOP-THAM0AM ~]
$ cd /mnt/g/CYS/3/ap/project
[sudhan@DESKTOP-THAM0AM ~]
$ ./privacy_analyzer
=====
Privacy Policy Analyzer
=====

[DatabaseManager] Default constructor called.
[KeywordMatcher] Ready to match privacy policy text.
[LLMManager] Initialized with model: gemma:2b
[TextAnalyzer] Ready to analyze privacy policy text.
[TextAnalyzer] LLM server connected successfully.

----- MENU -----
1. Load text from file
2. Enter text manually
3. Analyze privacy policy
4. Generate AI summary
5. Store current policy in database
6. Store analysis results in database
7. View stored policies
8. View analysis history
9. Exit
-----
Enter your choice: 1
Enter file name (e.g., policy.txt): file.txt
Loading file...
[TextAnalyzer] File loaded successfully: file.txt
File loaded successfully!

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-----
Enter your choice: 3
Analyzing...
[DatabaseManager] Connected to database successfully!
[DatabaseManager] Keywords fetched: 48
[KeywordMatcher] Loaded 48 keywords from DB.

Starting keyword analysis...

Analyzing Privacy Policy Text...
[collect] [collect] -> (data_collection)
[share] -> (data_sharing)
[opt-out] -> (user_rights)
[collect] [collect] -> (data_collection)
[share] -> (data_sharing)
[opt-out] -> (user_rights)
[access] [access] -> (user_rights)
[consent] [consent] [consent] -> (user_rights)
[protect] -> (security)
[third-party] -> (data_sharing)
[personal information] [personal information] [personal information] -> (data_collection)
[privacy policy] [privacy policy] -> (user_rights)

Summary of Detected Terms:
Category: data_collection | Occurrences: 3
Category: data_sharing | Occurrences: 3
Category: security | Occurrences: 1
Category: user_rights | Occurrences: 9
Analysis completed! Keyword data stored for AI summary.

----- MENU -----
```

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-----
Enter your choice: 4
Generating AI-powered summary...
Consulting AI...
Generating AI-powered summary based on keyword analysis...
This may take 10-20 seconds...
[LLMManager] Using simplified prompt for Gemma 2B
[LLMManager] Prompt length: 542
[LLMManager] Sending request to LLM (timeout: 120s)...
[LLMManager] Successfully generated summary

AI-Powered Privacy Policy Summary (Based on Keyword Analysis)
=====
Sure, here's a summary of the privacy policy in 3-4 sentences:

1. This policy outlines the types of personal data we collect from visitors to our website and how we use it.

2. We communicate with users, provide information about our programs, services, and activities, and process inquiries.

3. We also manage our volunteer and internship programs and conduct surveys and other research to gather insights.

4. We are committed to protecting user privacy and complying with all applicable laws and regulations.
=====
Note: This AI analysis is based on detected privacy-related keywords and should be verified by legal experts.

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-----
Enter your choice: 5
Storing policy in database...
[DatabaseManager] Policy table created/verified successfully.
[DatabaseManager] Policy stored successfully. Characters: 1965
[TextAnalyzer] Policy stored in database successfully.
Policy stored successfully in database!

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9. Exit
-----
Enter your choice: 6
Storing analysis results...
[DatabaseManager] Retrieved 3 stored policies.
[DatabaseManager] Analysis table created/verified successfully.
[DatabaseManager] Analysis results stored successfully for policy ID: 3
[TextAnalyzer] Analysis results stored successfully for policy ID: 3
Analysis results stored successfully!

----- MENU -----
1. Load text from file
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3. Analyze privacy policy
4. Generate AI summary
5. Store current policy in database
6. Store analysis results in database
```

```
sudhan@DESKTOP-THAM0AA x + v
Analysis results stored successfully!

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-----
Enter your choice: 7
[DatabaseManager] Retrieved 3 stored policies.

Stored Privacy Policies:
=====
ID: 3
Source: file (file.txt)
Characters: 1965
Date: 2025-10-16 23:29:40
Preview: 1. Information We Collect
We collect various types of personal information from individuals who vis...
[DatabaseManager] Retrieved 1 analysis results for policy ID: 3
Analyses: 1 time(s)
-----
ID: 2
Source: file (filee.txt)
Characters: 1342
Date: 2025-10-16 10:03:05
Preview: We value your privacy and are committed to protecting your personal information. Our website collect...
[DatabaseManager] Retrieved 1 analysis results for policy ID: 2
Analyses: 1 time(s)
-----
ID: 1
Source: file (filee.txt)
Characters: 1342
Date: 2025-10-15 21:23:58
Preview: We value your privacy and are committed to protecting your personal information. Our website collect...
[DatabaseManager] Retrieved 0 analysis results for policy ID: 1
Analyses: 0 time(s)
-----
```

```
sudhan@DESKTOP-THAM0AA x + v

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9. Exit
-----
Enter your choice: 8
Enter policy ID to view analysis history: 3
[DatabaseManager] Retrieved 1 analysis results for policy ID: 3

Analysis History for Policy ID: 3
=====
Analysis #1 - 2025-10-16 23:29:45
=====
Keyword Analysis:
KEYWORD ANALYSIS RESULTS:
=====

data_collection:
  Occurrences: 7
  Keywords found: collect(4x), personal information(3x)

data_sharing:
  Occurrences: 3
  Keywords found: share(2x), third-party

security:
  Occurrences: 1
  Keywords found: protect

user_rights:
  Occurrences: 9
  Keywords found: access(2x), consent(3x), opt-out(2x), privacy policy(2x)

----- MENU -----
1. Load text from file

Analysis History for Policy ID: 3
=====
Analysis #1 - 2025-10-16 23:29:45
=====
Keyword Analysis:
KEYWORD ANALYSIS RESULTS:
=====

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  Occurrences: 7
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-----
Enter your choice: 9
Exiting program. Goodbye!
[TextAnalyzer] Analysis completed and resources cleared.
[DatabaseManager] Connection closed.
[DatabaseManager] Destructor called, connection closed.

(sudhan@DESKTOP-THAM0AA) - [~/mnt/g/CYS/3/ap/project]
$
```

LLM used : Gemma

Database:

```
sudhan@DESKTOP-THAM0AM x + v
[sudhan@DESKTOP-THAM0AM ~]~/mnt/g/CYS/3/ap/project
$ sudo mysql -u root -p
[sudo] password for sudhan:
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 32
Server version: 11.8.3-MariaDB-1+b1 from Debian -- Please help get to 10k stars at https://github.com/MariaDB/Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> use privacy_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [privacy_db]> show tables;
+-----+
| Tables_in_privacy_db |
+-----+
| policy_analysis      |
| privacy_keywords     |
| stored_policies      |
+-----+
3 rows in set (0.001 sec)

MariaDB [privacy_db]> desc privacy_keywords;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id    | int(11) | NO | PRI | NULL | auto_increment |
| keyword | varchar(255) | NO | | NULL |
| category | varchar(100) | NO | | NULL |
+-----+-----+-----+-----+-----+
3 rows in set (0.014 sec)

MariaDB [privacy_db]> select * from privacy_keywords;
+-----+-----+-----+
| id | keyword | category |
+-----+-----+-----+
| 1 | collect | data_collection |
| 2 | share  | data_sharing  |
+-----+-----+-----+
```

```
sudhan@DESKTOP-THAM0AM x + v
MariaDB [privacy_db]> select * from privacy_keywords;
+-----+-----+-----+
| id | keyword | category |
+-----+-----+-----+
| 1 | collect | data_collection |
| 2 | share  | data_sharing  |
| 3 | cookie | data_collection |
| 4 | delete | user_rights   |
| 5 | opt-out | user_rights   |
| 6 | track  | data_collection |
| 7 | advertising | data_sharing  |
| 8 | collect | data_collection |
| 9 | gather | data_collection |
| 10 | store | data_collection |
| 11 | retain | data_collection |
| 12 | share  | data_sharing  |
| 13 | disclose | data_sharing  |
| 14 | transfer | data_sharing  |
| 15 | sell   | data_sharing  |
| 16 | cookie | tracking      |
| 17 | tracking | tracking      |
| 18 | log    | tracking      |
| 19 | analytics | tracking      |
| 20 | delete | user_rights   |
| 21 | opt-out | user_rights   |
| 22 | access | user_rights   |
| 23 | modify | user_rights   |
| 24 | consent | user_rights   |
| 25 | permission | user_rights   |
| 26 | encrypt | security      |
| 27 | secure | security      |
| 28 | protect | security      |
| 29 | breach | security      |
| 30 | anonymize | security      |
| 31 | mask   | security      |
| 32 | third-party | data_sharing  |
| 33 | marketing | data_sharing  |
| 34 | advertising | data_sharing  |
| 35 | personal information | data_collection |
| 36 | sensitive data | data_collection |
| 37 | profile | data_collection |
| 38 | preferences | data_collection |
| 39 | location | data_collection |
+-----+-----+-----+
```

```
sudhan@DESKTOP-THAM0AA x + v
35 | personal information | data_collection |
36 | sensitive data      | data_collection |
37 | profile             | data_collection |
38 | preferences         | data_collection |
39 | location            | data_collection |
40 | device info         | data_collection |
41 | IP address          | tracking         |
42 | cookies             | tracking         |
43 | user behavior       | tracking         |
44 | tracking pixel       | tracking         |
45 | data subject        | user_rights     |
46 | right to be forgotten | user_rights     |
47 | privacy policy      | user_rights     |
48 | security incident   | security        |
+-----+-----+-----+
48 rows in set (0.007 sec)

MariaDB [privacy_db]> desc stored_policies;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id     | int(11) | NO | PRI | NULL | auto_increment |
| content | text | NO | | NULL | |
| source | varchar(50) | NO | | NULL | |
| filename | varchar(255) | YES | | NULL | |
| char_count | int(11) | YES | | NULL | |
| analysis_date | timestamp | YES | | current_timestamp() | |
| created_at | timestamp | YES | | current_timestamp() | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.005 sec)

MariaDB [privacy_db]> desc policy_analysis;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id     | int(11) | NO | PRI | NULL | auto_increment |
| policy_id | int(11) | NO | MUL | NULL | |
| keyword_analysis | text | NO | | NULL | |
| ai_summary | text | YES | | NULL | |
| analysis_date | timestamp | YES | | current_timestamp() | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.003 sec)

MariaDB [privacy_db]>
```

Sample file used to test: file.txt

```
New Text Document (2).txt • main.cpp file.txt file.txt
File Edit View H1  B I A
1. Information We Collect
We collect various types of personal information from individuals who visit our Website, including but not limited to:

2. Use of Personal Data
We use the collected personal data for the following purposes:

To communicate with you about our initiatives, events, and updates
To provide information about our programs, services, and activities
To process and respond to your inquiries, feedback, or requests
To manage our volunteer and internship programs
To conduct research and analysis related to our NGO's focus areas
To collaborate with corporate partners and government bodies for community projects and research

3. Data Sharing and Consent
We value your privacy and are committed to maintaining the confidentiality of your personal data. We do not share your personal information with third parties without your explicit consent, except as required by law or as outlined in this Privacy Policy.

By submitting forms on our Website or participating in surveys and research activities conducted by us, you grant us the right to use the data for research and associated activities. Your data will be anonymized and aggregated whenever possible to ensure your privacy is protected.

4. Security Measures
We take reasonable steps to safeguard the personal information collected through our Website. We use appropriate technical, administrative, and physical measures to protect against unauthorized access, disclosure, alteration, or destruction of your personal data.

5. Your Rights
You have the right to:

Access and review the personal data we have collected about you
Request corrections or updates to your personal data
Withdraw your consent to the use of your data
Request the deletion of your personal data (subject to legal requirements)
Opt-out of receiving communications from us

6. Links to Third-Party Websites
This Privacy Policy is governed by and construed in accordance with the laws of India

Ln 30, Col 86 1,936 characters Plain text 100% Windows (CRLF) UTF-8
```

Class Diagram:

C++ Privacy Policy Analyzer UML

